NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-15733/43D
6 May 1994
SUPERSEDING
MIL-F-15733/43C
2 December 1982

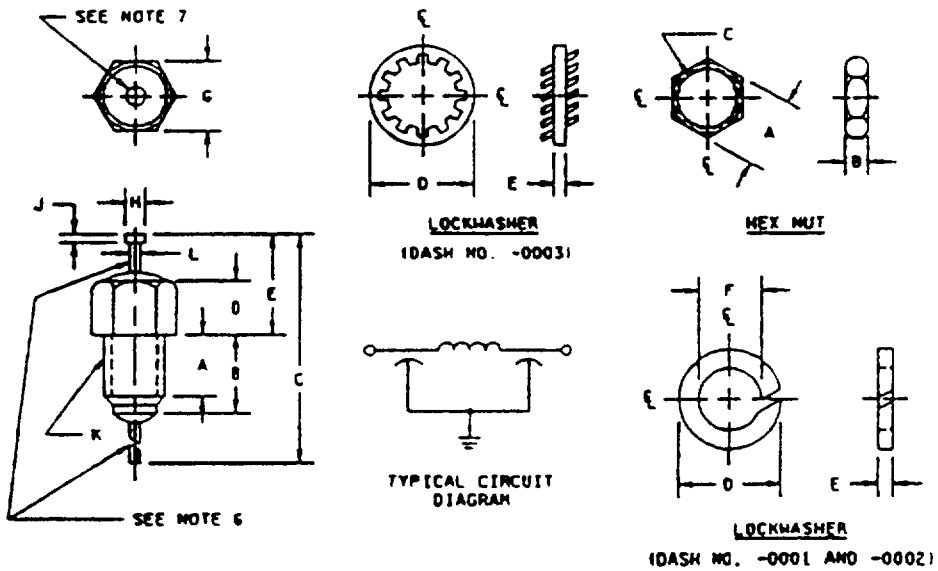
PERFORMANCE SPECIFICATION

FILTERS, RADIO FREQUENCY INTERFERENCE,
STYLE FL70

Part or Identifying Numbers (PIN's) M15733/43-0001 and M15733/43-0002 are inactive for new design after the date of this specification. For new design use M15733/61-0002 for M15733/43-0001 and M15733/61-0008 for M15733/43-0002.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-15733.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. Recommended mounting torque: 3 in-lb (-0001) and 6 in-lb (-0002 and -0003) maximum.
5. Mounting hardware (lockwasher and hex nut) will be supplied with each filter.
6. Leads shall be silver coated or solder coated copper.
7. Turret head is optional.

FIGURE 1. Case and circuit diagram

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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Filter dimensions.

Dash no.	Dimension									
	A		B		C		D		E	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
0001	.208 (5.28)	.228 (5.79)	.271 (6.88)	.291 (7.39)	.781 (19.84)	.843 (21.41)	.177 (4.50)	.197 (5.00)	.349 (8.86)	.369 (9.37)
0002	.206 (5.23)	.226 (5.74)	.256 (6.50)	.276 (7.01)	.859 (21.82)	.921 (23.39)	.083 (2.11)	.103 (2.62)	.281 (7.14)	.343 (8.71)
0003	.203 (5.16)	.233 (5.92)	.266 (6.76)	.296 (7.52)	.750 (19.05)	.874 (22.20)	.172 (4.37)	.202 (5.13)	.328 (8.33)	.390 (9.91)

Dash no.	Dimension								
	G		H		J		K	L	
	Min	Max	Min	Max	Min	Max	THD	Min	Max
0001	.240 (6.10)	.260 (6.60)	.050 (1.27)	.070 (1.78)	.010 (0.25)	.030 (0.76)	.216-28 UNF-2A	.027 (0.69)	.037 (0.94)
0002	.177 (4.50)	.197 (5.00)	.050 (1.27)	.070 (1.78)	.010 (0.25)	.030 (0.76)	.164-32 UNC-2A	.027 (0.69)	.037 (0.94)
0003	.235 (5.97)	.265 (6.73)	.050 (1.27)	.070 (1.78)	.010 (0.25)	.030 (0.76)	.216-32 UNEF-2A	.035 (0.89)	.045 (1.14)

Hardware dimensions.

Dash no.	Hex nut					Lockwasher					
	A		B		C	D		E		F	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max	Min	Max
0001	.240 (6.10)	.260 (6.60)	.070 (1.78)	.080 (2.03)	.216-28 UNF-2B	.351 (8.92)	.369 (9.37)	.045 (1.14)	.051 (1.30)	.219 (5.56)	.229 (5.82)
0002	.240 (6.10)	.260 (6.60)	.073 (1.85)	.078 (1.98)	.164-32 UNC-2B	.263 (6.68)	.273 (6.93)	.034 (0.86)	.038 (0.97)	.168 (4.27)	.176 (4.47)
0003	.302 (7.67)	.322 (8.18)	.070 (1.78)	.080 (2.03)	.216-32 UNEF-2B	.371 (9.42)	.381 (9.68)	.020 (0.51)	.024 (0.61)	N/A	N/A

FIGURE 1. Case and circuit diagram - Continued

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TABLE I. Electrical characteristics.

Dash number	Rated voltage		Capacitance pF minimum	Minimum insertion loss (dB) in accordance with MIL-STD-220 2/											
				+25°C						-55°C, +125°C					
	DC volts	AC 1/ rms volts		10 MHz	30 MHz	100 MHz	200 MHz	300 MHz	1-10 GHz	10 MHz	30 MHz	100 MHz	200 MHz	300 MHz	1-10 GHz
0001	200	140	1500	---	---	---	45	45	45	---	---	---	28	45	45
0002	100	70	1500	---	---	---	50	50	50	---	---	---	28	50	50
0003	500	---	2500	12	20	45	---	60	70	4	11	25	---	50	70

1/ DC to 400 Hz.

2/ Full-load insertion loss measurements shall be performed at 10 MHz. Measurements above this frequency shall be performed at no-load.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Metal.

Weight: 2.6 grams, maximum.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 10 amperes maximum.

Voltage drop: 0.10 volts maximum.

Voltage conditioning (quality conformance inspection only). The test shall be performed as follows:

Test temperature: 125°C.

Test voltage: 200 percent of the rated dc voltage. Charging current shall not exceed 50 mA.

Current load: Not applicable.

Points of application of test voltage: Between either terminal and case.

Duration of exposure to test voltage: 100 ±4 hours.

Seal: Not applicable.

Capacitance to ground: Method 305, MIL-STD-202.

Test frequency: 1,000 Hz ±100 Hz.

Test voltage: 0.5 V rms at 25°C.

Measured capacitance shall be as specified in table I

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Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage shall be three times the rated dc voltage applied for 1 to 5 seconds. Charging current shall not exceed 50 mA.

Barometric pressure (reduced): Method 105, test condition E, MIL-STD-202.

Dielectric withstanding voltage: In accordance with initial requirements, except that the test voltage shall be 120 percent of the rated dc voltage applied for 1 to 5 seconds.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Insulation resistance shall be measured at 25°C between either terminal and case, and shall be not less than 10,000 megohms.

Electrification voltage: 100 V dc after 2 minutes (maximum) electrification.

Insertion loss at +25°C and temperature extremes: See table 1.

Overload: In accordance with MIL-F-15733.

Measurements at 25°C after test:

Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A, MIL-STD-202.

Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion: Not applicable.

Shock (specified pulse): Method 213, test condition C, MIL-STD-202.

Electrical load: 100 percent of rated voltage between either terminal and case. After completion of the shock test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Vibration, high frequency: Method 204, test condition C, MIL-STD-202, except that part 2 shall be 20 g's. Duration shall be 3 hours each in two mutually perpendicular planes.

Electrical load: Rated voltage and rated current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: Not applicable.

Life: Method 108, test condition D, MIL-STD-202.

Test temperature 125°C.

Test voltage: 2 times rated dc voltage with no load at 125°C for 1,000 hours.

Measurements after test:

Dielectric withstanding voltage initial voltage applied for 1 to 5 seconds and shall meet initial requirements.

Insulation resistance shall be not less than 1,000 megohms.

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Capacitance to ground, and insertion loss (check test) shall be measured and shall meet the initial requirements.

Marking: Each filter shall be marked with the military PIN, as shown on figure 2. In addition, full marking in accordance with method I requirements of MIL-STD-1285 shall be marked on each unit package.

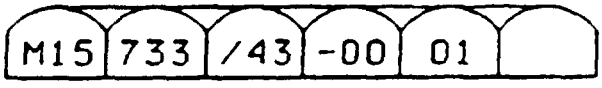


FIGURE 2. Hex flats-expanded view, part marking.

PIN: M15733/43- (dash number from table I and figure 1).

Application note: These nonhermetically sealed filters may be susceptible to moisture intrusion when subjected to repeated thermal cycling. If these items are to be utilized in applications enduring harsh environments, the user should consider placing them within hermetic enclosures.

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: In accordance with MIL-F-15733.

Quality conformance inspection: In accordance with MIL-F-15733 except as follows:

Group A inspection for all dash numbers shall be in accordance with table II. Lots having greater than 10 percent rejects shall be considered reject lots. Reject lots, at the option of the manufacturer, may be reworked and submitted to group A inspection of table II.

TABLE II. Group A inspection.

Test or inspection	Sampling procedure
Voltage conditioning	100%
Capacitance to ground	100%
Dielectric withstanding voltage	100%
Insulation resistance	100%
Voltage drop 1/	100%
Insertion loss (check test)	100%
Visual and mechanical inspection	See MIL-F-15733, group A inspection

1/ DC voltage drop/dc resistance test for dc rated filters shall be performed on a sample basis in accordance with MIL-F-15733, group A inspection.

Note: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - ES

(Project 5915-0377-01)

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